

Using Machine Learning To Predict Cancer Drug Response

SECTION: K61

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Objectives

1. **Explore** cancer genomics and drug response datasets from GDSC.
2. **Preprocess** and prepare gene expression data for analysis.
3. **Build** machine learning models for drug response prediction.
4. **Evaluate** model performance using regression metrics.
5. **Identify** important genomic predictors associated with drug sensitivity.
6. **Validate** prediction results and interpret biological significance.

1. Introduction

Cancer is one of the most common and challenging diseases worldwide, and one of the main difficulties in cancer treatment is that patients and cancer cells do not always respond to drugs in the same way. Some cancer cells may be highly sensitive to a treatment, while others become resistant. Because of this, researchers have become increasingly interested in using genomic data and machine learning techniques to better understand drug response and improve personalized medicine approaches.

In this project, gene expression and drug response data from the Genomics of Drug Sensitivity in Cancer (GDSC) database were used to predict the sensitivity of cancer cell lines to drugs targeting the Receptor Tyrosine Kinase (RTK) signaling pathway. The workflow included data collection, preprocessing, feature reduction, exploratory data analysis, machine learning model development, evaluation, deep learning analysis, and biological interpretation. Since the original dataset was very large, the number of genes was reduced to the top 10,000 most variable genes to make the analysis more manageable while still keeping biologically informative features.

2. Materials and Methods

2.1 Dataset Collection and Preparation

Gene expression and drug response datasets were downloaded from the Genomics of Drug Sensitivity in Cancer (GDSC) database through the Sanger Institute DepMap resources. Initial exploration and preprocessing were performed in Jupyter Notebook using Python with the imports of pandas and NumPy libraries. The gene expression dataset initially contained expression values for more than 41,000 genes across approximately 1,900 cancer cell lines, with genes arranged as rows and cell lines as columns. The expression matrix was transposed into a compatible format, with each row representing a cancer cell line sample identified by SANGER_MODEL_ID and each column representing a gene expression feature.

In addition, the drug response dataset contained information on anti-cancer drug sensitivity, including drug identifiers, pathway information, and LN_IC50 values. LN_IC50 values were used as the prediction target for machine learning analysis. The structure and dimensions of both datasets were examined before integration. Drug response records associated with the RTK signaling pathway were selected from the drug response dataset. Variance filtering was applied to the gene expression dataset to retain the top 10,000 most variable genes. Finally, the reduced expression dataset was merged with the RTK-specific drug response dataset using SANGER_MODEL_ID as a shared identifier to generate the final integrated dataset.

2. Materials and Methods

2.2 Data Cleaning and Preprocessing

The dataset was cleaned and preprocessed before model training to improve data quality and ensure compatibility with machine learning algorithms. The preprocessing pipeline included missing value handling, duplicate checking, data type validation, feature selection, and outlier analysis.

Missing values were identified across the gene expression dataset by calculating the total number and percentage of null values in each feature. Columns containing more than 50% missing values were removed from the dataset. The remaining missing numerical values were imputed using the median strategy because the median is less sensitive to extreme values compared to the mean.

Duplicate rows were checked using the `duplicated()` function to avoid repeated samples that could negatively affect model performance.

Data type validation was performed to ensure that numerical gene expression features were stored in machine-learning-compatible formats.

Gene expression columns were selected by removing metadata-related columns such as sample identifiers, drug names, pathway information, and target variables. This step ensured that only relevant gene expression features were used during model training.

Outlier analysis was performed using boxplot visualizations and the Interquartile Range (IQR) statistical method. The target variable (LN_IC50) and a subset of gene expression features were analyzed to identify potential extreme values.

The IQR method was calculated using:

$$\text{IQR} = Q3 - Q1$$

The lower and upper outlier boundaries were determined using:

$$\text{Lower Bound} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Upper Bound} = Q3 + 1.5 \times \text{IQR}$$

Outliers were retained because extreme drug response values may contain biologically significant information related to cancer sensitivity and resistance mechanisms.

2. Materials and Methods

2.3 Feature Engineering And Preprocessing

Performing feature engineering and processing on the cleaned genomic dataset “Processed_Data” to prepare the data for machine learning modeling. The target variable (LN_IC50) was separated from the genomic features. Non-biological identifier columns, including DRUG_ID, were handled separately to avoid misleading numerical interpretation during model training. Gene expression features were standardized using StandardScaler, which transformed all genomic features to mean = 0, standard deviation = 1. This normalization reduced differences in feature magnitude and improved compatibility with machine learning algorithms. Since DRUG_ID represents categorical drug identities rather than continuous numerical values, One-Hot Encoding was applied. Each drug was converted into separate binary columns (0/1), allowing the models to treat drugs as independent categories. The scaled genomic features and encoded drug features were then combined into a final normalized feature matrix ready for predictive modeling.

2. Materials and Methods

2.4 Exploratory Data Analysis (EDA)

The dataset used in this project contains drug response and gene expression data collected from multiple cancer cell lines, and data preprocessing was performed to clean and prepare the dataset for analysis. Numerical columns related to gene expression and LN_IC50 values were selected for exploratory data analysis (EDA).

Several visualization techniques were applied, including histograms, bar plots, and correlation analysis. Histograms were used to study the distribution patterns of numerical features such as LN_IC50, VIM, KRT19, SPARC, and RPL36A-HNRNPH2.

Variance analysis was performed to identify the top 10 most variable genes across samples. In addition, correlation analysis was conducted to determine the genes most positively and negatively associated with LN_IC50 values.

2. Materials and Methods

2.5 Model Development

The objective of this stage was to develop and compare regression models for predicting drug response. The target variable was LN_IC50, which is a continuous numerical value. Therefore, the task was treated as a regression problem rather than a classification problem.

The normalized genomic feature matrix was used as the input dataset. It contained 10,297 samples and 10,010 columns in total.

The target column was LN_IC50, while the remaining 10,009 columns were used as input features. Rows containing missing values were removed before training to ensure that the models could run without errors.

Value	Data Component
10297	Total Samples
10009	Input Features
LN_IC50	Target Variable
8237	Training Sample
2060	Testing Samples

After preprocessing, the dataset was divided into the feature matrix X and the target vector y . The feature matrix contained all input variables, while the target vector contained the LN_IC50 values. The data was then split into training and testing sets using an 80/20 split. The training set contained 8,237 samples, and the test set contained 2,060 samples. Four regression models were trained and compared: Linear Regression, Ridge Regression, Random Forest, and XGBoost. Linear Regression was used as a baseline model. Ridge Regression was used as a regularized linear model, which is useful for high-dimensional data. Random Forest and XGBoost were included as nonlinear ensemble models.

The models were evaluated using RMSE, MAE, and R^2 . RMSE and MAE measure prediction error, where lower values indicate better performance. R^2 measures how much variation in the target variable is explained by the model, where higher values indicate better performance.

2. Materials and Methods

2.6 Model Evaluation

In this part of the project, model evaluation and external validation were performed to assess the predictive performance and generalization ability of the trained machine learning models. Multiple models were compared using evaluation metrics including RMSE, MAE, and R2 score. To further evaluate model generalization, an independent external genomic dataset obtained from the DepMap/CCLL database was used. After preprocessing and feature matching, 8654 shared genomic features were successfully identified between the external dataset and the trained model features. The external data were then applied to the trained Ridge Regression model to generate drug response predictions. Several visualization techniques, including scatter plots, distribution plots, and residual-like analysis, were used to evaluate model behavior on the external dataset. In addition, 5-fold cross-validation was performed to assess model stability and predictive consistency

2. Materials and Methods

2.7 Deep Learning (PyTorch)

Deep learning was applied to the predictive framework developed to model the relationship between high-dimensional genomic signatures and pharmacological response. The dataset comprised 10,010 input features, consisting of baseline transcriptomic profiles (mRNA expression levels of genes such as VIM, KRT8, and SPARC) and one-hot encoded drug identifiers.

The model was constructed through a multi-layer Deep Neural Network(DNN) with three hidden layers to capture complex, non-linear biological interactions. Rectified Linear Unit (ReLU) activation functions were employed to ensure model generalization and mitigate overfitting, given the high feature-to-sample ratio. Dropout regularization ($p=0.3$) was integrated between layers. The model has optimized over 100 epochs using a Mean Squared Error (MSE) loss function to regress the target variable of the natural logarithm of the half-maximal inhibitory concentration (LN_IC50).

Data type validation was performed to ensure that numerical gene expression features were stored in machine-learning-compatible formats.

Gene expression columns were selected by removing metadata-related columns such as sample identifiers, drug names, pathway information, and target variables. This step ensured that only relevant gene expression features were used during model training.

3. Results

3.1 Dataset Preperation and Integration Results

Initial exploration showed that the gene expression and drug response datasets had different structures and dimensions. The gene expression dataset contained more than 41,000 gene expression features across approximately 1,900 cancer cell lines.

After preprocessing, the expression matrix was successfully transposed into a machine-compliant format. Filtering of the drug response dataset produced an RTK signaling-specific subset containing targeted therapy response records.

Variance filtering reduced the number of gene expression features from more than 41,000 genes to 10,000 genes. The reduced expression dataset was then merged successfully with the RTK drug response dataset using SANGER_MODEL_ID as the common identifier.

The final integrated dataset contained cancer cell line identifiers, RTK pathway drug information, LN_IC50 drug sensitivity values, and 10,000 gene expression features for downstream machine learning analysis.

3. Results

3.2 Data Cleaning and Preprocessing Results

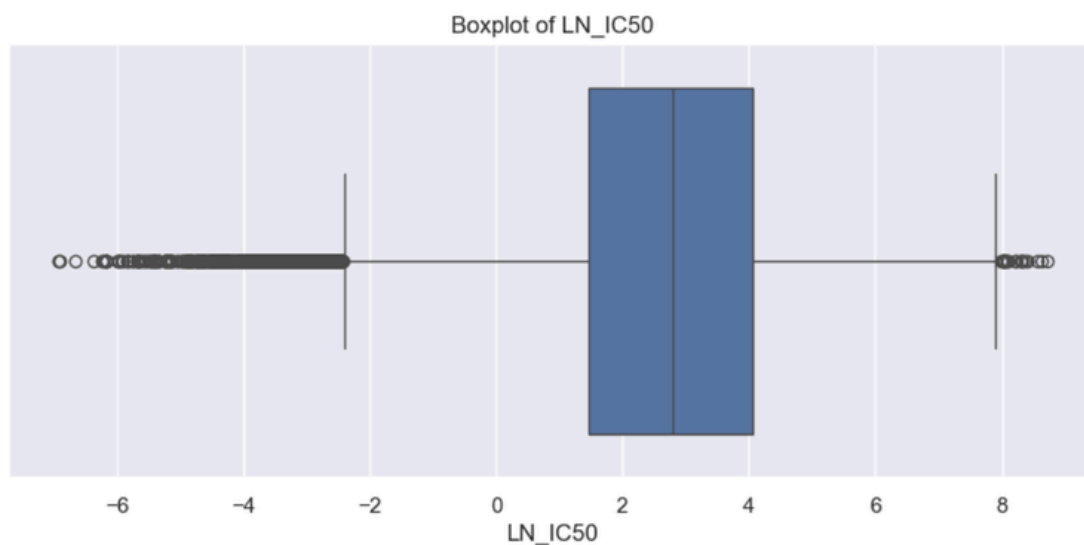
The preprocessing stage successfully improved the quality and consistency of the dataset before model training. Missing value analysis identified several features containing null values.

Columns with more than 50% missing data were removed, while the remaining missing numerical values were handled using median imputation.

Duplicate checking was performed to identify repeated samples within the dataset. No duplicated rows were detected after inspection.

Feature selection reduced the dataset to relevant gene expression features by excluding metadata-related columns such as identifiers, drug names, and pathway information. This step prepared the dataset for machine learning analysis.

Outlier analysis was conducted using boxplot visualizations and the Interquartile Range (IQR) method.



Several extreme values were identified in the LN_IC50 target variable and gene expression features. However, these outliers were retained because they may represent biologically meaningful variations associated with drug sensitivity and resistance. The final processed dataset was clean, structured, and suitable for machine learning model development. A clean and ready to process data file is then exported as “Processed_Data”.

3. Results

3.3 Feature Engineering And Preprocessing

The preprocessing pipeline successfully generated a normalized feature matrix containing:

- scaled genomic expression features
- One-Hot Encoded drug categories.

The final dataset contained:

- 10,297 samples
- 10,009 processed features

After scaling, genomic feature values became standardized and centered around zero, improving feature comparability across the dataset. The One-Hot Encoding process converted drug identifiers into binary categorical columns, such as:

- DRUG_1029
- DRUG_1034
- DRUG_1049

This prevented false numerical relationships between drug identifiers while preserving drug-specific information. The final processed dataset was saved as

“Normalized_Genomic_Feature_Matrix.csv” and prepared for downstream regression modeling. Discussion

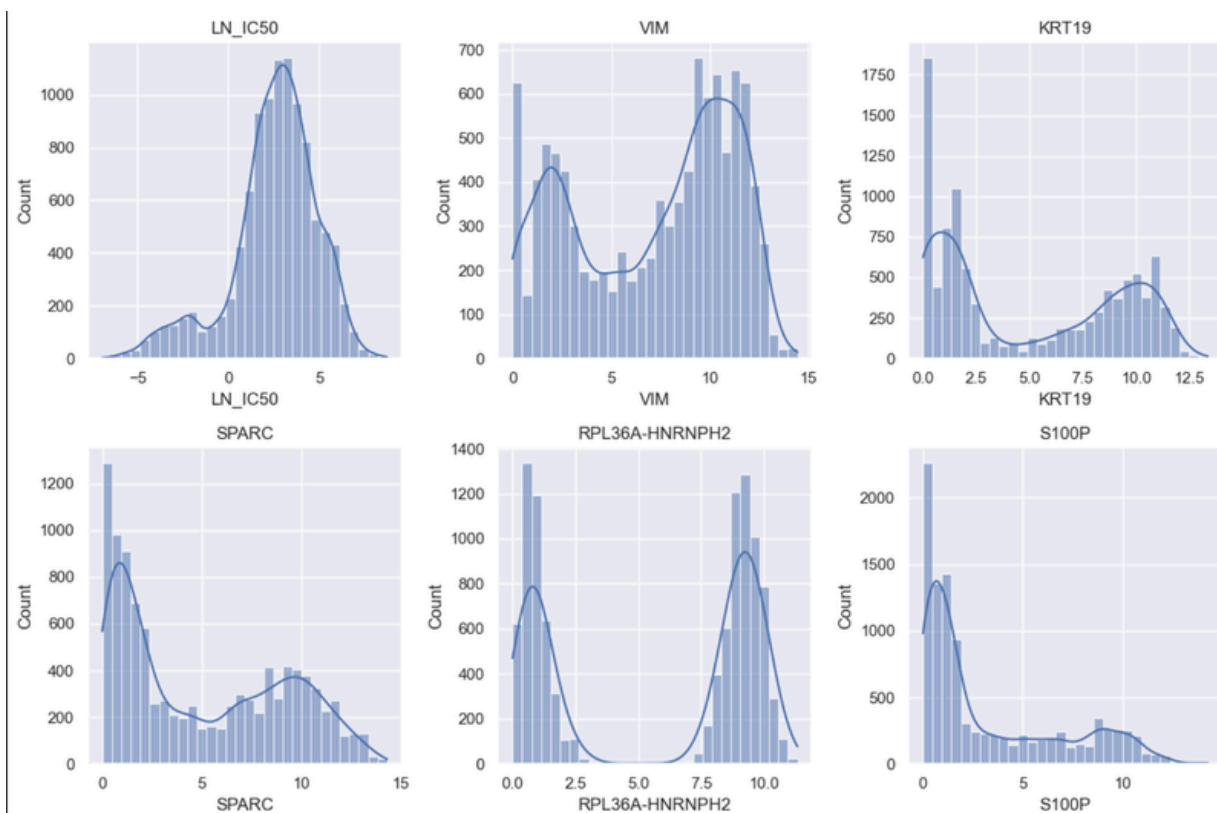
3. Results

3.4 Exploratory Data Analysis (EDA)

The exploratory data analysis revealed that the dataset contains 10,297 samples and 10,002 features. Drug distribution analysis showed that Staurosporine was the most frequently occurring drug in the dataset, followed by several other commonly tested drugs.

Histogram analysis demonstrated that numerical features exhibited different distribution patterns, including normal, skewed, and bimodal distributions. This indicates variability in gene expression across samples.

Variance analysis identified KRT19 as the gene with the highest variance, followed by RPL36A-HNRNPH2, SPARC, and VIM. Correlation analysis showed that genes such as ITGA3, SDC4, and PLPP2 were positively correlated with LN_IC50, while WAS, PTPN7, and AGAP2 showed negative correlations.



3. Results

3.5 Model Development and Evaluation

Ridge Regression has proven to achieve the best overall performance out of multiple models, resulting in the lowest Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) values and the highest coefficient of determination (R^2) score. The best selected alpha value for Ridge Regression was 0.1; although Linear regression showed similar performance, it didn't exceed Ridge Regression across all three metrics.

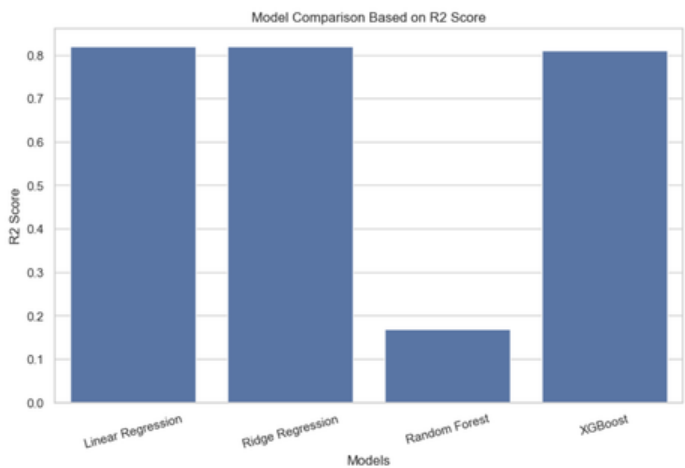
Other models, such as XGBoost, have shown great performance as well, with an R^2 score of 0.8111, but they didn't outperform Ridge Regression. The weakest performance was shown in Random Forest, with the highest prediction error and the lowest R^2 score.

Therefore, Ridge Regression was selected as the best performing model due to its good scores on the test results

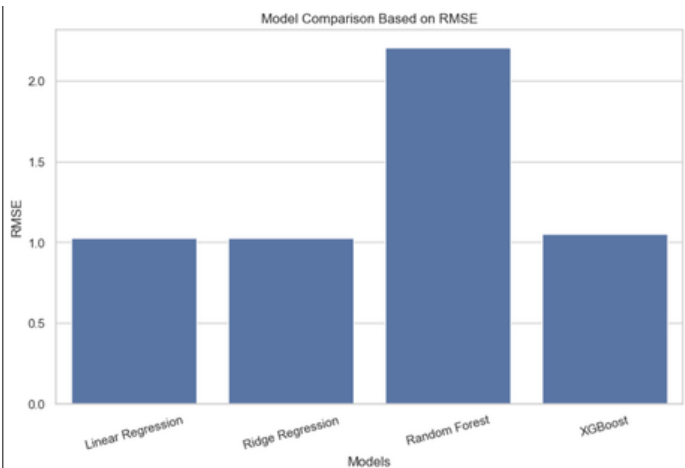
Model’s performance demonstration

R ²	Test MAE	Test RMSE	Model
0.8200	0.7776	1.0276	Linear Regression
0.8206	0.7755	1.0261	Ridge Regression
0.1689	1.6130	2.2082	Random Forest
0.8111	0.8044	1.0529	XGBoost

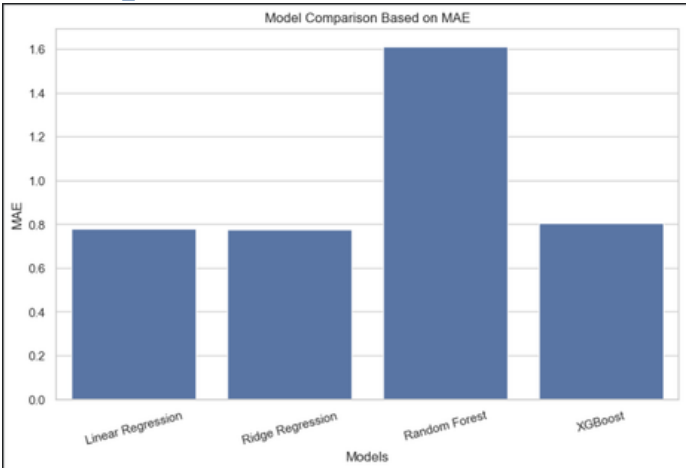
Comparison Based on R²



Comparison Based on RMSE



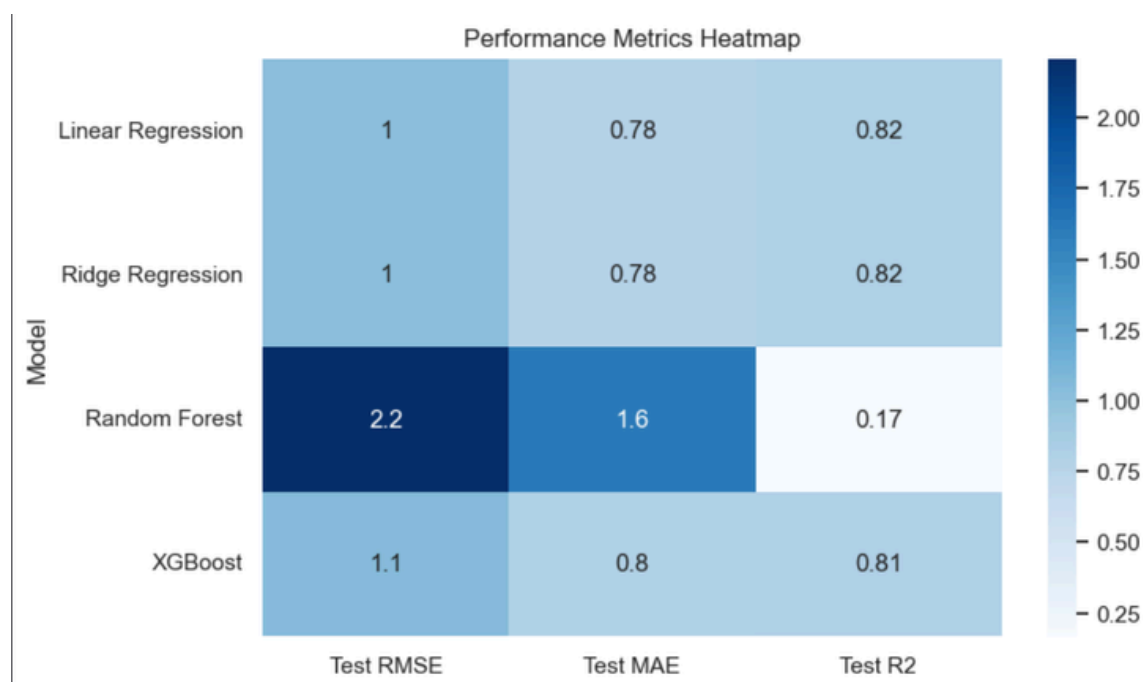
Comparison Based on MAE



3. Results

3.5 Model Development and Evaluation

Among the evaluated machine learning models, the Ridge Regression model demonstrated the best overall performance based on RMSE, MAE, and R2 metrics. The model successfully generated diverse prediction values on the external genomic dataset after feature matching. The external validation analysis identified 8654 shared genomic features between the trained model and the independent external dataset. Furthermore, the 5-fold cross-validation produced a mean validation score of approximately 0.67, indicating moderate but relatively stable predictive performance across validation folds.



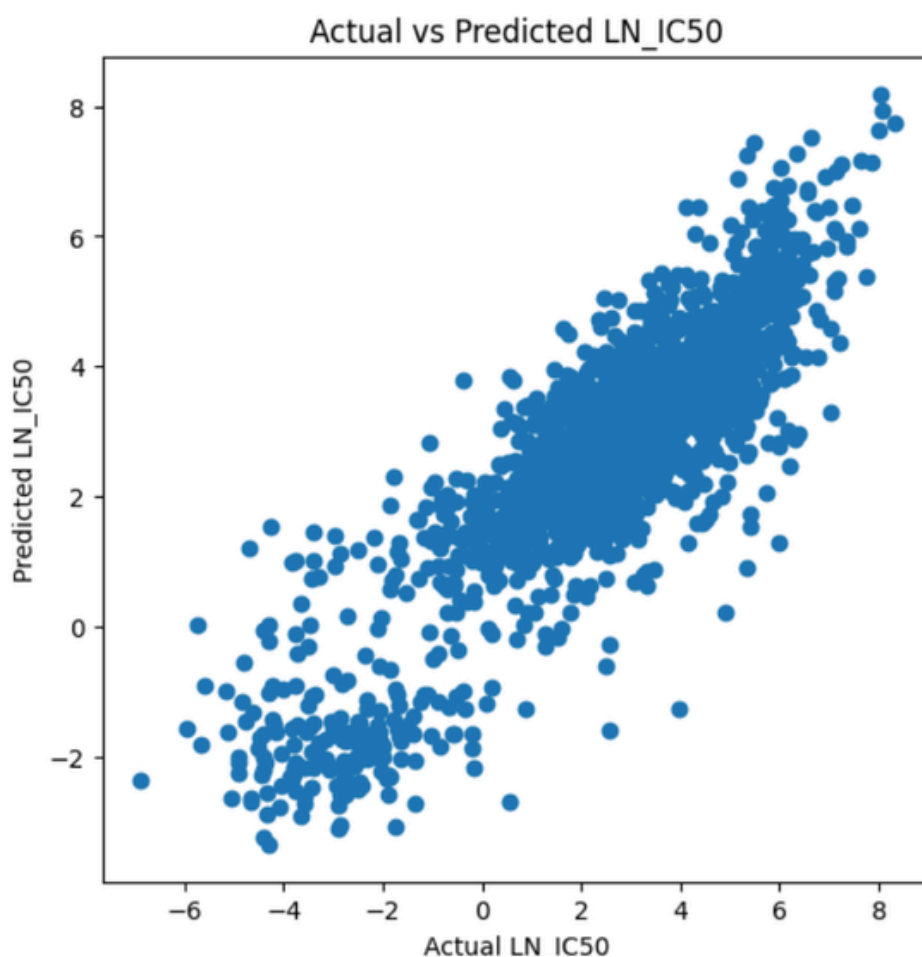
3. Results

3.6 Deep Learning Using PyTorch Results

The model demonstrated robust predictive performance on the unseen test partition, achieving a coefficient of determination (R^2) of 0.7258 and a Root Mean Squared Error (RMSE) of 1.2685. This indicates that the architecture successfully accounts for approximately 72.6% of the variance in drug sensitivity.

The analysis of feature importance had revealed that chemical identity was the dominant predictor. Features DRUG_1029 and DRUG_1034 exhibited the highest importance scores, significantly outweighing individual genomic features.

The "Actual vs. Predicted" regression plot confirms a strong linear concordance across a wide dynamic range of IC50 values



4. Discussion

4.1 Importance of Dataset Preparation for Machine Learning Analysis

Preprocessing and integration of the datasets were necessary before machine learning analysis because the original datasets were large and computationally intensive. Reducing the number of genes and selecting only RTK pathway drugs produced a more manageable dataset for downstream analysis.

Focusing specifically on RTK signaling pathway drugs also made the dataset more pathway-specific. As a result, the developed models may become more specialized for predicting responses to RTK-targeted therapies rather than generalizing to drugs associated with unrelated pathways. However, limiting the dataset to a single pathway may reduce biological variability and improve consistency between drug response records.

Variance filtering reduced dataset dimensionality while retaining highly variable genes that may contain more informative expression patterns across cancer cell lines. The final integrated dataset provided a suitable foundation for downstream machine learning, deep learning, and biological interpretation analyses related to cancer drug response prediction.

4. Discussion

4.2 Data cleaning and preprocessing Discussion

Data cleaning and preprocessing played an essential role in improving the overall quality and reliability of the dataset before model development. Handling missing values helped reduce noise and minimize the negative impact of incomplete data on machine learning performance. The use of median imputation provided a robust approach for replacing missing numerical values while reducing sensitivity to extreme observations.

Duplicate checking ensured data consistency by preventing repeated samples from introducing bias into the learning process. In addition, validating data types and selecting relevant gene expression features improved compatibility with machine learning algorithms and reduced unnecessary information within the dataset.

Outlier analysis revealed the presence of extreme values in both the LN_IC50 target variable and gene expression features. Although outliers are commonly removed in some preprocessing pipelines, they were intentionally retained in this study because extreme drug response values may contain biologically significant patterns related to cancer sensitivity and resistance mechanisms. Preserving these observations may help improve the model's ability to capture complex biological relationships within the data.

4. Discussion

4.3 Feature Engineering And Preprocessing

The preprocessing strategy improved the quality and reliability of the genomic dataset for machine learning applications. Feature scaling was particularly important because genomic expression values originally existed on highly variable numerical ranges.

Without normalization, features with large magnitudes could dominate model learning and introduce numerical bias.

The use of One-Hot Encoding for DRUG_ID preserved categorical drug information while preventing machine learning models from interpreting drug identifiers as continuous values. This improved the biological validity of the dataset and reduced misleading statistical relationships.

Combining normalized genomic features with encoded drug categories created a balanced feature space that allowed downstream regression models to learn both:

- genomic expression patterns
- drug-specific effects

Overall, the preprocessing pipeline produced a clean, structured, and machine-learning-ready dataset suitable for predictive drug response analysis.

4. Discussion

4.4 Exploratory Data Analysis (EDA)

The results indicate that gene expression patterns vary significantly among cancer cell lines, which may influence drug response behavior.

Highly variable genes are important because they contain useful biological information and may improve predictive modeling performance. The correlation analysis highlighted several genes that may play a role in drug sensitivity and resistance. Positive correlations suggest direct relationships with LN_IC50 values, whereas negative correlations indicate inverse relationships.

Overall, the exploratory data analysis provided valuable insights into the dataset structure, feature variability, and gene-drug relationships. These findings can support future machine learning models for drug response prediction and biomarker discovery.

4. Discussion

4.5 Model Development And Evaluation

The results show that the linear models performed strongly on this dataset. Both Linear Regression and Ridge Regression achieved R^2 values of approximately 0.82, meaning that they explained about 82% of the variation in LN_IC50.

Ridge Regression slightly outperformed Linear Regression.

Although the improvement was small, Ridge was preferred because it includes regularization. This is useful when working with a large number of input features, as it helps improve model stability and reduce the effect of overly large coefficients.

XGBoost also performed well but did not provide better results than Ridge Regression in this experiment. Random Forest had the weakest performance, which may be due to the high dimensionality of the dataset and the lightweight tuning used to reduce computational cost.

Although the model performance on the external dataset was moderate rather than exceptionally high, the relatively stable cross-validation results suggest that the model maintained consistent predictive behavior and did not completely overfit the original training database. Duplicate checking ensured data consistency by preventing repeated samples from introducing bias into the learning process.

4. Discussion

4.5 Model Development And Evaluation

In addition, validating data types and selecting relevant gene expression features improved compatibility with machine learning algorithms and reduced unnecessary information within the dataset. Overall, Ridge Regression was selected as the final model for this stage because it achieved the lowest prediction error and the highest explanatory performance on the test set. Its saved model and feature names can also be used in the next stage for further evaluation and biological interpretation.

4. Discussion

4.6 Deep Learning Using PyTorch

The results show that drug response depends on two main factors:

- Strength of the drug
- Biology of the cell.

The high importance of drug IDs like DRUG_1029 and DRUG_1034 suggests that the chemical's power is the primary driver of the result. The model uses these IDs to set a baseline for how effective the treatment will be across all cells.

However, the strong R^2 score of 0.726 proves that the genes are essential for accuracy. Markers like VIM and KRT8 describe the cell's state (its EMT status), which tells the model if a cell is naturally tough or easy to kill. In simple terms, the drug identity determines the general impact zone, while the 10,000+ genes provide the biological details needed to fine-tune the final prediction. This confirms that while the drug type starts the reaction, the cell's genetic profile decides the final level of sensitivity or resistance.

5. Feature Importance and Biological Interpretation

Feature Importance

Machine Learning

Feature importance analysis from Ridge Regression and XGBoost identified both RTK-targeting drugs and transcriptomic biomarkers as important predictors of LN_IC50. Ridge Regression achieved the best overall performance and was therefore used as the primary model for biological interpretation. Important RTK inhibitors included Motesanib, Staurosporine, PD173074, Foretinib, and AZD1332, suggesting that drug-specific effects strongly influenced cancer cell sensitivity patterns. Important genes such as VIM, KRT19, SPARC, S100P, UCHL1, and LCN2 were associated with epithelial–mesenchymal transition (EMT), tumor progression, invasion, and treatment resistance mechanisms. Similar predictors were also observed in XGBoost, supporting the biological relevance of these findings. Overall, the machine learning models successfully integrated transcriptomic and pharmacogenomic information to identify biologically meaningful predictors associated with RTK-targeted drug response.

Deep Learning

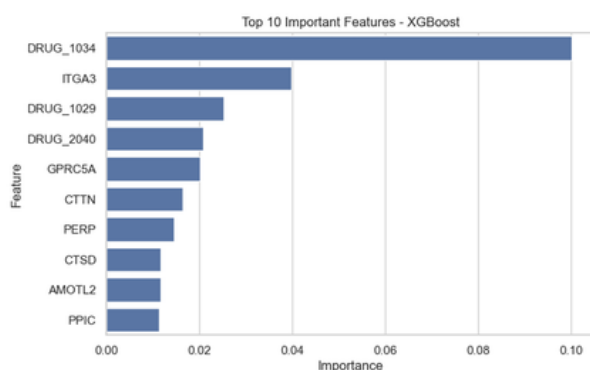
The deep learning model also achieved strong predictive performance and identified both RTK-targeting drugs and cancer-associated transcriptomic features as important contributors to LN_IC50 prediction. Important RTK inhibitors included Motesanib, Staurosporine, PD173074, and Crizotinib, while important genes included VIM, KRT8, KRT19, S100P, LCN2, and UCHL1. These genes are associated with EMT, tumor progression, invasion, and drug resistance mechanisms. The results suggest that the deep learning model captured complex nonlinear relationships between RTK-targeted therapies and transcriptomic patterns linked to cancer drug sensitivity and resistance. Although the identified features were biologically relevant, feature importance reflects predictive contribution rather than direct biological causation.

Feature Importance

Machine Learning

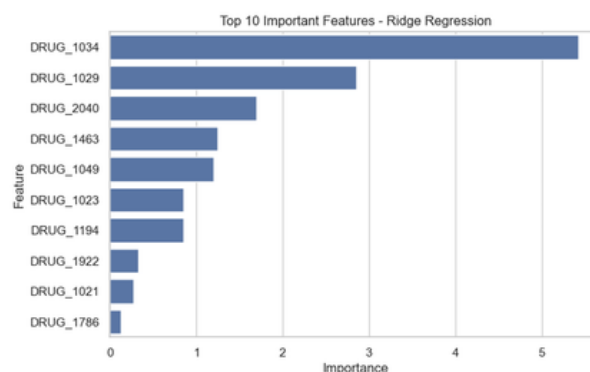
Top 10 Important Features - XGBoost

	Feature	Importance
10000	DRUG_1034	0.100110
187	ITGA3	0.039835
9999	DRUG_1029	0.025187
10008	DRUG_2040	0.020800
122	GPRC5A	0.020145
1310	CTTN	0.016407
340	PERP	0.014515
2479	CTSD	0.011678
558	AMOTL2	0.011551
819	PPIC	0.011255



Top 10 Important Features - Ridge Regression

	Feature	Importance
10000	DRUG_1034	5.424043
9999	DRUG_1029	2.854597
10008	DRUG_2040	1.697024
10004	DRUG_1463	1.244876
10001	DRUG_1049	1.196776
9998	DRUG_1023	0.854708
10003	DRUG_1194	0.850726
10006	DRUG_1922	0.329079
9997	DRUG_1021	0.269818
10005	DRUG_1786	0.127855



Deep Learning

DRUG_ID DRUG_NAME

0	1021	Axitinib
936	1023	GW441756
1864	1029	Motesanib
2789	1034	Staurosporine
3732	1049	PD173074
4674	1083	Crizotinib
5615	1194	SB505124
5872	1463	AZD1332
6570	1786	AZD4547
7504	1922	Cediranib
8437	1936	Savolitinib
9369	2040	Foretinib

	Feature	Importance
9999	DRUG_1029	10.819161
10000	DRUG_1034	9.894964
10001	DRUG_1049	5.180589
10004	DRUG_1463	5.122636
9998	DRUG_1023	3.790335
10008	DRUG_2040	3.618149
10003	DRUG_1194	3.101679
9997	DRUG_1021	2.181860
10005	DRUG_1786	1.825873
10007	DRUG_1936	1.736247

Biological Interpretation

The model had achieved a strong predictive model, indicating that gene expression and drug related features can effectively predict cancer drug response[1]. Each anticancer drug differs in its mechanisms of action and target pathways; therefore, drug-specific variables have contributed to the prediction[2]. The analysis also highlighted biologically important genes such as VIM and KRT8, which are associated with epithelial–mesenchymal transition (EMT). EMT is linked to cancer progression, metastasis, and drug resistance, suggesting that resistant cancer cells may have more aggressive molecular characteristics. The analysis also highlighted biologically important genes such as VIM and KRT8, which are associated with epithelial–mesenchymal transition (EMT). EMT is linked to cancer progression, metastasis, and drug resistance, suggesting that resistant cancer cells may have more aggressive molecular characteristics[3].

These findings indicate that the model captured biologically meaningful patterns rather than only statistical associations.

Overall, the results support the use of machine learning in precision oncology, where transcriptomic data can help predict therapeutic response and identify potential mechanisms of resistance. However, further validation using clinical datasets is necessary before applying these findings in patient treatment.

5. References

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